from this prior, we guess that all three have a normalized price of zero, and we have varying levels of uncertainty regarding our guesses. Further, as we are working with a random distribution, we could draw samples from this multivariate Gaussian. These samples are shown as connected faded diamonds.

2.2.2 Updating multivariate Gaussian distributions

With a multivariate Gaussian distribution in hand, we will see how we can update this distribution given some data that we observe in this subsection. Specifically, following our example at the beginning of the chapter, we'd like to derive the *posterior belief* about these prices upon observing the value of either B or C. This is an important task, as it is how a multivariate Gaussian, as well as a Gaussian process, learns from data.

IMPORTANT **Conditioning in Bayesian probability**

This update process is sometimes called conditioning: deriving the conditional distribution of a variable, given that we know the value of some other variable. More specifically, we are conditioning our belief that is the joint tri-variate Gaussian on either B or C, obtaining the joint posterior distribution for these three variables.

Here, using the Bayes' theorem in Figure 2.2, we can derive this posterior distribution in closed form. However, the derivation is rather math-heavy, so we won't be going into it here. We just need to know that we have a formula in which we can plug the value of B or C that we'd like to condition on, and the formula will tell us what the posterior distributions of A, B, and C are, specifically their updated means and variances. (Later in the chapter, we will see that when we implement a Gaussian process in Python, GPyTorch will take care of this math-heavy update for us.)

SIDEBAR **More Gaussian process math**

For the interested reader, this formula and its derivation can be found in Chapter 2, Section 2 of the Gaussian Processes for Machine Learning book (the MIT Press, 2006), which is often considered the bible of Gaussian processes.

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